CHAPTER 5

THE NATURAL NUMBERS

Our fundamental intuitive understanding of the natural numbers is that there is a (least) number 0, that every number n has a successor Sn, and that if we start with 0 and construct in sequence the successor of every number

$$0,\ S0 = 1,\ S1 = 2,\ S2 = 3,\ \ldots$$

forever, then in time we will reach every natural number. In set theoretic terms we can capture this intuition by the following axiomatic characterization.

5.1. Definition. A **Peano system** or **system of natural numbers** is any structured set

$$(\mathbb{N}, 0, S) = (\mathbb{N}, (0, S))$$

which satisfies the following conditions.

1. $\mathbb{N}$ is a set which contains the element 0, $0 \in \mathbb{N}$.
2. S is a function on $\mathbb{N}$, $S : \mathbb{N} \to \mathbb{N}$.
3. S is an injection, $Sn = Sm \Longrightarrow n = m$.
4. For each $n \in \mathbb{N}$, $Sn \neq 0$.
5. **Induction Principle.** For each $X \subseteq \mathbb{N}$,

$$[0 \in X \;\&\; (\forall n \in \mathbb{N})[n \in X \Longrightarrow Sn \in X]] \Longrightarrow X = \mathbb{N}.$$

These obvious properties of the natural numbers are called the **axioms of Peano** in honor of the Italian logician and mathematician who first proposed them as an axiomatic foundation of number theory, following their earlier formulation by Dedekind. Most significant among them is the Induction Principle, whose typical application is illustrated in the proof of the next lemma.

5.2. Lemma. *In a Peano system* $(\mathbb{N}, 0, S)$, *every element* $n \neq 0$ *is a successor,*

$$\textit{if } n \neq 0, \textit{ then } (\exists m \in \mathbb{N})[n = Sm],$$

and for each n, $Sn \neq n$.

Proof. To prove the first assertion by the Induction Principle, it is enough to show that the set

$$X = \{n \in \mathbb{N} \mid n = 0 \vee (\exists m \in \mathbb{N})[n = Sm]\}$$

satisfies the conditions

$$0 \in X, \quad (\forall n \in \mathbb{N})[n \in X \Longrightarrow Sn \in X],$$

and both of these are obvious from the definition of X. In the same way, for the second assertion it is enough to verify that $S0 \neq 0$ (which holds because, in general, $Sn \neq 0$) and that $Sn \neq n \Longrightarrow SSn \neq Sn$: this holds because S is one-to-one, so that $SSn = Sn \Longrightarrow Sn = n$. ⊣

Number theory is one of the richest and most sophisticated fields of mathematics and it is by no means obvious that it can be developed on the basis of these five, simple properties; in fact they do not suffice, one also needs to use set theory which (in its naive form) Peano took for granted, as part of "logic". Here we will only show that the axioms imply the first, most basic properties of addition, multiplication and the ordering on the natural numbers, which is all we need. The proofs we will give, however, are characteristic examples of the use of the Peano axioms in the more advanced parts of the theory of numbers.

If number theory can be developed from the Peano axioms, then to give a faithful representation of the natural numbers in set theory, it is enough to prove from the axioms the following two theorems.

5.3. Theorem (Existence of the natural numbers). *There exists at least one Peano system* $(\mathbb{N}, 0, S)$.

5.4. Theorem (Uniqueness of the natural numbers). *For any two Peano systems* $(\mathbb{N}_1, 0_1, S_1)$ *and* $(\mathbb{N}_2, 0_2, S_2)$, *there exists one* (*and only one*) *bijection*

$$\pi : \mathbb{N}_1 \rightarrowtail\!\!\!\!\!\twoheadrightarrow \mathbb{N}_2$$

such that

$$\begin{aligned} \pi(0_1) &= 0_2, \\ \pi(S_1 n) &= S_2 \pi(n) \quad (n \in \mathbb{N}_1). \end{aligned}$$

A bijection π which satisfies these identities is an **isomorphism** of the two systems $(\mathbb{N}_1, 0_1, S_1)$ and $(\mathbb{N}_2, 0_2, S_2)$, so that the theorem asserts that *any two Peano systems are* (*uniquely*) *isomorphic*.

The Existence Theorem is very simple and we can prove it immediately.

5.5. Proof of the existence of natural numbers, **5.3**. The Axiom of Infinity (**VI**) guarantees the existence of a set I such that

$$\begin{aligned} &\emptyset \in I, \\ &(\forall n)[n \in I \Longrightarrow \{n\} \in I]. \end{aligned}$$

Using this I, first we define the family of sets

$$\mathscr{I} = \{X \subseteq I \mid \emptyset \in X \;\&\; (\forall n)[n \in X \Longrightarrow \{n\} \in X]\}$$

so that obviously $I \in \mathscr{I}$, and then we set

$$\mathbb{N} = \bigcap \mathscr{I}, \quad 0 = \emptyset, \quad S = \{(n, m) \in \mathbb{N} \times \mathbb{N} \mid m = \{n\}\}.$$

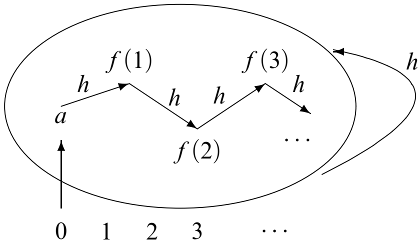

FIGURE **5.1**. The Recursion Theorem.

To finish off the proof, it suffices to verify that this triple $(\mathbb{N}, 0, S)$ is a Peano system. To begin with, $\mathbb{N} \in \mathscr{I}$, because $X \in \mathscr{I} \Longrightarrow \emptyset \in X$ and hence $\emptyset \in \bigcap \mathscr{I} = \mathbb{N}$, and by the same thinking,

$$n \in \mathbb{N} \Longrightarrow (\forall X \in \mathscr{I})[n \in X] \Longrightarrow (\forall X \in \mathscr{I})[\{n\} \in X] \Longrightarrow \{n\} \in \mathbb{N}.$$

This implies immediately the first two of the Peano axioms, the next two hold because (in general, for all n, m) $\{n\} = \{m\} \Longrightarrow n = m$ and $\{n\} \neq \emptyset$, and the Induction Principle follows directly from the definition of $\mathbb{N}$ as an intersection. ⊣

To prove the Uniqueness Theorem **5.4**, we need the next fundamental result of axiomatic number theory.

5.6. Recursion Theorem. *Assume that $(\mathbb{N}, 0, S)$ is a Peano system, E is some set, $a \in E$, and $h : E \to E$ is some function: it follows that there exists exactly one function $f : \mathbb{N} \to E$ which satisfies the identities*

$$f(0) = a,$$
$$f(Sn) = h(f(n)) \quad (n \in \mathbb{N}).$$

The Recursion Theorem justifies the usual way by which we define functions on the natural numbers, **by recursion**[12] (or induction): to define $f : \mathbb{N} \to E$, first we specify the value $f(0) = a$ and then we supply a function $h : E \to E$ which determines the value $f(Sn)$ of f at every successor Sn from the value $f(n)$ at its predecessor n, $f(Sn) = h(f(n))$. Our basic intuition about the natural numbers with which we started this chapter clearly justifies such definitions, so we should also be able to justify them on the basis of the axioms.

Before we establish the Recursion Theorem, let us use it in the next proof which is a typical example of the way it is applied.

5.7. Proof of the uniqueness of the natural numbers, **5.4**. We assume that $(\mathbb{N}_1, 0_1, S_1)$ and $(\mathbb{N}_2, 0_2, S_2)$ are Peano systems. *By the Recursion Theorem on*

[12]The terms "recursion" and "induction" are often used synonymously in mathematics. We will follow the more recent convention which distinguishes **recursive definitions** from **inductive proofs**.

$(\mathbb{N}_1, 0_1, S_1)$ with $E = \mathbb{N}_2$, $a = 0_2$, $h = S_2$, there exists exactly one function $\pi : \mathbb{N}_1 \to \mathbb{N}_2$ which satisfies the identities

$$\pi(0_1) = 0_2,$$
$$\pi(S_1 n) = S_2\pi(n) \quad (n \in \mathbb{N}_1),$$

and it suffices to verify that this π is a (one-to-one) correspondence.

(1) π **is a surjection,** $\pi : \mathbb{N}_1 \twoheadrightarrow \mathbb{N}_2$. Obviously $0_2 \in \pi[\mathbb{N}_1]$ since $0_2 = \pi(0_1)$, and

$$\begin{aligned} m \in \pi[\mathbb{N}_1] &\Longrightarrow (\exists n \in \mathbb{N}_1)[m = \pi(n)] \\ &\Longrightarrow (\exists n \in \mathbb{N}_1)[S_2 m = S_2\pi(n) = \pi(S_1 n)] \\ &\Longrightarrow S_2 m \in \pi[\mathbb{N}_1], \end{aligned}$$

so that by *the Induction Principle on* $(\mathbb{N}_2, 0_2, S_2)$, $\pi[\mathbb{N}_1] = \mathbb{N}_2$.

(2) π **is an injection**, $\pi : \mathbb{N}_1 \rightarrowtail \mathbb{N}_2$. It suffices to verify that if we set

$$X = \{n \in \mathbb{N}_1 \mid (\forall m \in \mathbb{N}_1)[\pi(m) = \pi(n) \Longrightarrow m = n]\},$$

then

$$0_1 \in X, \quad n \in X \Longrightarrow S_1 n \in X,$$

since together with *the Induction Principle on* $(\mathbb{N}_1, 0_1, S_1)$, this implies that $X = \mathbb{N}_1$, which means that π is an injection. For the first condition,

$$\begin{aligned} m \neq 0_1 &\Longrightarrow m = S_1 m' \text{ for some } m', \text{ by Lemma } \mathbf{5.2} \\ &\Longrightarrow \pi(m) = \pi(S_1 m') = S_2\pi(m') \neq 0_2, \end{aligned}$$

so that if $\pi(m) = \pi(0_1) = 0_2$, then $m = 0_1$ and $0_1 \in X$. For the second condition, it is enough to show that

$$n \in X \,\&\, \pi(m) = \pi(S_1 n) \Longrightarrow m = S_1 n.$$

By the hypothesis

$$\pi(m) = \pi(S_1 n) = S_2\pi(n) \neq 0_2,$$

which implies that $m \neq 0_1$, since $\pi(0_1) = 0_2$ and $0_1 \in X$. By Lemma **5.2** again, $m = S_1 m'$ for some $m' \in \mathbb{N}_1$,

$$\pi(m) = \pi(S_1 m') = S_2\pi(m')$$

and the hypothesis $\pi(m) = \pi(S_1 n)$ yields

$$S_2\pi(m') = S_2\pi(n),$$

which implies $\pi(m') = \pi(n)$. This, in turn, implies $m' = n$ because $n \in X$, so that $m = S_1 m' = S_1 n$, the required conclusion. ⊣

5.8. Proof of the Recursion Theorem. Assume the hypotheses and define first the set $\mathscr{A}$ of all *approximations* of the function which we want to construct:

$$\begin{aligned} p \in \mathscr{A} \iff_{\mathrm{df}} & \text{Function}(p) \\ & \& \,\text{Domain}(p) \subseteq \mathbb{N} \,\&\, \text{Image}(p) \subseteq E \\ & \& \, 0 \in \text{Domain}(p) \,\&\, p(0) = a \\ & \& \, (\forall n \in \mathbb{N})[Sn \in \text{Domain}(p) \\ & \qquad \qquad \Longrightarrow n \in \text{Domain}(p) \,\&\, p(Sn) = h(p(n))]. \end{aligned} \tag{5-1}$$

In words, each $p \in \mathscr{A}$ is a function with domain a subset of $\mathbb{N}$ and values in E; by the third clause, Domain(p) contains 0; and by the last one, Domain(p) is "closed downward", i.e., if $Sn \in$ Domain(p), then $n \in$ Domain(p), and the value $p(Sn)$ is determined by $p(n)$. Some examples of approximations are

$$\{(0,a)\}, \{(0,a),(S0,f(a))\}, \{(0,a),(S0,f(a)),(SS0,f(f(a)))\}, \ldots$$

and they suggest how the required function f is "built up" stage-by-stage by the recursive definition. To prove the theorem, we need to show (from the axioms only, without " . . . " or "etc.") that there is exactly one approximation with domain of definition all of $\mathbb{N}$.

Lemma. *For all $p, q \in \mathscr{A}$ and $n \in \mathbb{N}$,*

$$n \in \text{Domain}(p) \cap \text{Domain}(q) \Longrightarrow p(n) = q(n).$$

Proof. The set

$$X = \{n \in \mathbb{N} \mid (\forall p, q \in \mathscr{A})\Big[n \in \text{Domain}(p) \cap \text{Domain}(q) \Longrightarrow p(n) = q(n)\Big]\}$$

clearly contains 0, since every $p \in \mathscr{A}$ satisfies $p(0) = a$. If

$$n \in X \,\&\, p \in \mathscr{A} \,\&\, q \in \mathscr{A} \,\&\, Sn \in \text{Domain}(p) \cap \text{Domain}(q),$$

then

$$\begin{aligned} p(Sn) &= h(p(n)) && \text{because } p \in \mathscr{A}, \\ &= h(q(n)) && \text{because } p(n) = q(n), \\ &= q(Sn) && \text{because } q \in \mathscr{A}, \end{aligned}$$

so that if $n \in X$, then $Sn \in X$. It follows by the Induction Principle that $X = \mathbb{N}$, which completes the proof. ⊣ (Lemma)

The Lemma implies immediately that *at most one* function $f : \mathbb{N} \to E$ belongs to $\mathscr{A}$, so to complete the proof of the theorem we need only show that *at least one* such f exists. This is the union

$$f = \bigcup \mathscr{A} = \{(n,w) \mid (\exists p \in \mathscr{A})[n \in \text{Domain}(p) \,\&\, p(n) = w]\},$$

which is a function, because

$$\begin{aligned} (n,w) \in f \,\&\, (n,w') \in f &\Longrightarrow (\exists p, q \in \mathscr{A})[(n,w) \in p \,\&\, (n,w') \in q] \\ &\Longrightarrow w = w' \text{ from the Lemma,} \end{aligned}$$

and then the definition of $\mathscr{A}$ and a similar calculation shows that $f \in \mathscr{A}$. The only thing left is to verify that $\text{Domain}(f) = \mathbb{N}$, and for that we will use once more the Induction Principle. To begin with, $0 \in \text{Domain}(f)$, since $0 \in \text{Domain}(p)$ for every $p \in \mathscr{A}$ and $\mathscr{A} \neq \emptyset$. If $n \in \text{Domain}(f)$, then there exists some function $p \in \mathscr{A}$ with $n \in \text{Domain}(p)$, and hence (easily)

$$q = p \cup \{(Sn, h(p(n)))\} \in \mathscr{A},$$

so that $Sn \in \text{Domain}(q) \subseteq \text{Domain}(f)$. $\dashv$

5.9. The natural numbers. We now fix a specific Peano system $(\mathbb{N}, 0, S)$ whose members we will henceforth call **natural numbers**, or just **numbers**, when there cannot be any confusion. Following Cantor, we denote the cardinal number of $\mathbb{N}$ by the first Hebrew letter with the subscript 0, pronounced "aleph-zero",

$$\aleph_0 =_{\text{df}} |\mathbb{N}|. \tag{5-2}$$

Later we will meet its followers $\aleph_1$, $\aleph_2$, etc. Functions $a : \mathbb{N} \to A$ with domain $\mathbb{N}$ are called (infinite) **sequences** and we often write their argument as a subscript,

$$a_n = a(n) \quad (n \in \mathbb{N}, a : \mathbb{N} \to A).$$

An obvious choice for $\mathbb{N}$ would be the system which we constructed in the proof of the Existence Theorem **5.3**, where $0 = \emptyset$,

$$\mathbb{N} = \{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \dots\}$$

and $Sn = \{n\}$. Another choice which some would prefer on philosophical grounds is to assert that there exists, in fact, a set

$$\mathbb{N} = \{0, 1, 2, \dots\}$$

of the "true natural numbers", which are not sets, and the successor function S is nothing like the artificial $(n \mapsto \{n\})$, but it is the "natural successor function" which associates with each number n "the next number" Sn. Zermelo's theory allows such non-sets (like the "true numbers") as atoms, and requires only one thing: that the system of natural numbers satisfies the Peano axioms, something which every serious person will surely grant. As far as the mathematical theory of numbers and sets is concerned, these two (and all other) choices of the objects we will call *numbers* are equivalent, since we will base all our proofs on the Peano axioms alone.

5.10. The Schröder-Bernstein Theorem 2.26. At this point, we should review the proof of this important result and verify that *we can now derive it from the axioms*; this is because the recursive definitions of the sequences of sets $\{A_n\}_{n\in\mathbb{N}}$ and $\{B_n\}_{n\in\mathbb{N}}$ are justified by the Recursion Theorem, and their basic properties are established by induction and simple manipulations of functions, which can all be based on the results in Chapter **4**.

There are many reformulations of the Recursion Theorem which are useful in applications. We state here just two of them, whose proofs illustrate how the theorem is used. Two more such results are included in the problems, **x5.20** and **x5.21**.

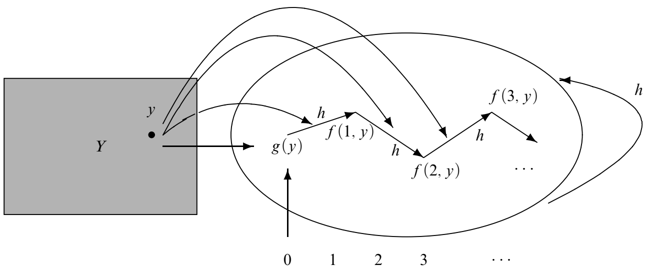

FIGURE **5.2**. The Recursion Theorem with parameters.

5.11. Corollary (Recursion with parameters). *For any two sets Y, E and functions*

$$g : Y \to E, \quad h : E \times Y \to E,$$

there exists exactly one function $f : \mathbb{N} \times Y \to E$ which satisfies the identities

$$\begin{aligned} f(0, y) &= g(y) && (y \in Y), \\ f(Sn, y) &= h(f(n, y), y) && (y \in Y, n \in \mathbb{N}). \end{aligned}$$

PROOF. For each $y \in Y$, we define the function $h_y : E \to E$ by the formula

$$h_y(w) = h(w, y),$$

and by the Recursion Theorem we know that there exists *exactly one* function

$$f_y : \mathbb{N} \to E$$

which satisfies the identity

$$\begin{aligned} f_y(0) &= g(y), \\ f_y(Sn) &= h_y(f_y(n)) = h(f_y(n), y). \end{aligned}$$

It follows immediately that the function $f : \mathbb{N} \times Y \to E$ defined by the formula

$$f(n, y) =_{\mathrm{df}} f_y(n) \quad (y \in Y, n \in \mathbb{N})$$

satisfies the conclusion of the Corollary. ⊣

5.12. Corollary (Recursion with the argument as parameter). *For every set E, each $a \in E$, and every function $h : E \times \mathbb{N} \to E$, there is exactly one function $f : \mathbb{N} \to E$ which satisfies the identities*

$$\begin{aligned} f(0) &= a, \\ f(Sn) &= h(f(n), n). \end{aligned}$$

Similarly, with parameters, for every

$$g : Y \to E, \quad h : E \times \mathbb{N} \times Y \to E,$$

there exists exactly one function $f : \mathbb{N} \times Y \to E$ *which satisfies the identities*

$$\begin{aligned} f(0, y) &= g(y) && (y \in Y), \\ f(Sn, y) &= h(f(n, y), n, y) && (y \in Y,\ n \in \mathbb{N}). \end{aligned}$$

PROOF. For the version with parameters, we define first a function

$$\phi : \mathbb{N} \times Y \to \mathbb{N} \times E$$

by recursion with parameters **5.11**, where the component functions First and Second are those of Exercise **4.3**:

$$\begin{aligned} \phi(0, y) &= (0, g(y)) \\ \phi(Sn, y) &= (S\mathrm{First}(\phi(n, y)), h(\mathrm{Second}(\phi(n, y)), \mathrm{First}(\phi(n, y)), y)). \end{aligned}$$

By induction on n, immediately,

$$\mathrm{First}(\phi(n, y)) = n,$$

and so the function ϕ satisfies the equations

$$\phi(0, y) = (0, g(y)), \quad \phi(Sn, y) = (Sn, h(\mathrm{Second}(\phi(n, y)), n, y)).$$

This implies immediately that the function

$$f(n, y) = \mathrm{Second}(\phi(n, y))$$

satisfies the required identities.

The uniqueness is easy to prove directly by induction on n. ⊣

We now use these results to define and establish the basic properties of addition, multiplication and the ordering on $\mathbb{N}$.

5.13. Addition and multiplication. The addition function on the natural numbers is defined by the recursion

$$\begin{aligned} n + 0 &= n, \\ n + (Sm) &= S(n + m), \end{aligned} \tag{5-3}$$

and multiplication is defined next, using addition, by the recursion

$$\begin{aligned} n \cdot 0 &= 0, \\ n \cdot Sm &= (n \cdot m) + n. \end{aligned} \tag{5-4}$$

In more detail, we know from **5.11** that there exists exactly one function $f : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ which satisfies the identities

$$\begin{aligned} f(0, n) &= g(n), \\ f(Sm, n) &= h(f(m, n), n), \end{aligned}$$

where the functions g and h have been given as sets of pairs,

$$\begin{aligned} g &= \{(n, n) \in \mathbb{N} \times \mathbb{N} \mid n \in \mathbb{N}\}, \\ h &= \{((z, n), w) \in (\mathbb{N} \times \mathbb{N}) \times \mathbb{N} \mid w = Sz\}, \end{aligned}$$

and we define addition by

$$n + m =_{\mathrm{df}} f(m, n),$$

i.e., $+ = \{((n,m),w) \mid ((m,n),w) \in f\}$. Such scholastic details do not enhance understanding (rather the opposite) and we will avoid them in the future.

5.14. Theorem. *Addition is* **associative**, *i.e., it satisfies the identity*

$$(n+m)+k = n+(m+k) \tag{5-5}$$

PROOF. First for $k = 0$,

$$(n+m)+0 = n+m = n+(m+0),$$

using twice the identity $w + 0 = w$ directly from the definition of addition. Inductively, assuming that for some k

$$(n+m)+k = n+(m+k), \tag{5-6}$$

we compute:

$$\begin{aligned}(n+m)+Sk &= S((n+m)+k)\\ &= S(n+(m+k)) \text{ by (5-6)}\\ &= n+S(m+k)\\ &= n+(m+Sk)\end{aligned}$$

where the steps we did not justify follow from the definition of addition. ⊣

The commutativity of addition is not quite so simple and requires two lemmas.

5.15. Lemma. *For every natural number n, $0+n = n$.*

PROOF. By induction, $0+0 = 0$ follows from the definition, and if $0+n = n$, then $0+Sn = S(0+n) = Sn$. ⊣

5.16. Lemma. *For all n, m, $n+Sm = Sn+m$.*

PROOF. By induction on m, first for $m = 0$, immediately from the definition:

$$n+S0 = S(n+0) = Sn = Sn+0.$$

At the induction step, we assume that for some m

$$n+Sm = Sn+m \tag{5-7}$$

and we must show that

$$n+SSm = Sn+Sm.$$

Compute:

$$\begin{aligned}n+SSm &= S(n+Sm) && \text{by the definition}\\ &= S(Sn+m) && \text{from (5-7)}\\ &= Sn+Sm && \text{by the definition.}\end{aligned}$$ ⊣

5.17. Theorem. *Addition is a* **commutative** *function, i.e., it satisfies the identity*

$$n+m = m+n.$$

PROOF is by induction on m, the basis being immediate from Lemma **5.15**. At the induction step, we assume that for some specific m

$$n + m = m + n \tag{5-8}$$

and we compute:

$$\begin{aligned} n + Sm &= S(n+m) && \text{by the definition} \\ &= S(m+n) && \text{from (5-8)} \\ &= m + Sn && \text{by the definition} \\ &= Sm + n && \text{by Lemma } \textbf{\textit{5.16}}. \end{aligned}$$ ⊣

5.18. Exercise. *For every natural number n, the function $(s \mapsto n+s)$ is one-to-one, so that $n + s = n + t \Longrightarrow s = t$, and in particular*

$$\textit{if } \ n + s = n, \ \textit{then } \ s = 0.$$

5.19. Definition. A binary relation $\leq$ on a set P is a **partial ordering** if it is reflexive, transitive and antisymmetric, i.e., for all $x, y, z \in P$,

$$\begin{aligned} & x \leq x && \text{(reflexivity)}, \\ & x \leq y \;\&\; y \leq z \Longrightarrow x \leq z && \text{(transitivity)}, \\ & x \leq y \;\&\; y \leq x \Longrightarrow x = y, && \text{(antisymmetry)}. \end{aligned}$$

In connection with partial orderings we will also use the notation

$$x < y \iff_{\text{df}} x \leq y \;\&\; x \neq y.$$

The partial ordering $\leq$ is **total**, or **linear**, or simply an **ordering**, if, in addition, any two elements of P are **comparable** in $\leq$, i.e.,

$$(\forall x, y \in P)[x \leq y \vee y \leq x],$$

or equivalently

$$(\forall x, y \in P)[x < y \vee x = y \vee y < x].$$

5.20. Definition. The binary relation $\leq$ on P is a **wellordering** of P, if it is a total ordering of P and, in addition, every non-empty subset of P has a least element,

$$(\forall X \subseteq P)[X \neq \emptyset \Longrightarrow (\exists x \in X)(\forall y \in X)[x \leq y]].$$

Correct English would have us call these "good orderings", and in fact this is what they are called in every other language, but the awkward "wellordering" has been established so firmly that it is hopeless to try and change it.

5.21. Definition. The order relation $\leq$ on the natural numbers is defined by the equivalence

$$n \leq m \iff_{\text{df}} (\exists s)[n + s = m].$$

The most basic property of $\leq$ is:

5.22. Lemma. *For all natural numbers n, m,*

$$n \leq Sm \iff n \leq m \vee n = Sm.$$

PROOF. If $n \le Sm$, then there is some t such that $n + t = Sm$ by the definition, and we consider two cases. CASE (1), $t = 0$. Now $n + 0 = Sm$, hence $n = Sm$. CASE (2), $n + t = Sm$ for some $t \neq 0$. Now, by Lemma **5.2**, $t = Ss$ for some s, so that $n + Ss = Sm$, hence $S(n + s) = Sm$, hence $n + s = m$ because S is an injection and hence $n \le m$. The converse direction of the Lemma is easier. ⊣

5.23. Theorem. *The relation $\le$ on the natural numbers is a wellordering.*

PROOF. Reflexivity is immediate from $n + 0 = n$ and transitivity holds because $n + s = m \,\&\, m + t = k \Longrightarrow n + (s + t) = k$. To prove antisymmetry, notice that if $n + s = m$ and $m + t = n$, then $n + (s + t) = n$ and by Exercise **5.18** we have $s + t = 0$; this implies $t = 0$ (otherwise $s + t$ is a successor) and hence $m = n$.

Proof of linearity. We show that $(\forall n)[n \le m \vee m \le n]$, by induction on m. Notice first that for every n, $n \le Sn$, because $n + S0 = Sn$.

BASIS. For every n, $0 + n = n$ and hence $0 \le n$.

INDUCTION STEP. We assume the induction hypothesis

$$(\forall n)[n \le m \vee m \le n]$$

and show that for each n, $n \le Sm \vee Sm \le n$. The induction hypotheses naturally splits the proof up in two cases. If $n \le m$, then $n \le Sm$ because $m \le Sm$ and $\le$ is transitive. If $m \le n$, then for some t, $m + t = n$, and again we have two cases: if $t = 0$, then $n = m \le Sm$, and if $t \neq 0$, then $t = Ss$ for some s, so $m + Ss = n$ and from Lemma **5.16** $Sm + s = n$, hence $Sm \le n$.

Proof of the wellordering property. Towards a contradiction, suppose that X is non-empty but has no least element and let

$$Y = \{n \in \mathbb{N} \mid (\forall m \le n)[m \notin X]\},$$

so that obviously

$$Y \cap X = \emptyset. \tag{5-9}$$

It is enough to show that $0 \in Y$ and $n \in Y \Longrightarrow Sn \in Y$, because then $Y = \mathbb{N}$ by the Induction Principle and hence $X = \emptyset$ by (5-9), which is a contradiction.

BASIS. $0 \in Y$. Since 0 is the least number, we must have $0 \notin X$ (otherwise X would have a least member) and also $m \le 0 \Longrightarrow m = 0 \Longrightarrow m \notin X$, so $0 \in Y$.

INDUCTION STEP. The induction hypothesis $n \in Y$ and the definition of Y imply that $(\forall m \le n) m \notin X$, and then we know from Lemma **5.22** that $m \le Sn \Longrightarrow m \le n \vee m = Sn$. Hence to verify that $Sn \in Y$, it is enough to show that $Sn \notin X$. But if Sn were a member of X, then it would be the least member of X since

$$\begin{aligned} m < Sn &\iff m \le n && \text{by Lemma } \mathbf{5.22} \\ &\Longrightarrow m \notin X && \text{by the ind. hyp.} \end{aligned}$$

This shows that $\le$ has the wellordering property and completes the proof of the theorem. ⊣

About wellorderings, in general, we will say a lot in Chapter **7**. In the special case of the natural numbers, the fact that $\mathbb{N}$ is well ordered by $\leq$ is another manifestation of the Induction Principle.

Before we begin studying the applications of recursion to the theory of finite and countable sets, we should recall the warning issued in **3.24**: some of them require the Axiom of Choice and we will not be able to justify them axiomatically until we add that axiom to our system in Chapter **8**. Most, however, can be established by judicious applications of the general method of proof which can be symbolized by the coupling

$$\text{recursive definition} \ - \ \text{inductive proof}.$$

First we repeat some of the definitions of Chapter **2**, with the axiomatic notions now at our disposal.

5.24. Definition. For any two natural numbers $n \leq m$, the (half-open) **interval** from n to m is the set

$$[n, m) =_{\text{df}} \{k \in \mathbb{N} \mid n \leq k \,\&\, k < m\}.$$

5.25. Exercise. *For each n, $[n, n) = \emptyset$, and for all $n \leq m$,*

$$[n, Sm) = [n, m) \cup \{m\}.$$

5.26. Definition. A set A is **finite** if there exists some natural number n such that $A =_c [0, n)$; **infinite** if it is not finite; and **countable** if it is finite or equinumerous with $\mathbb{N}$. By Proposition **2.7** (which follows easily from the axioms),

$$A \text{ is countable} \iff A \leq_c \mathbb{N}.$$

The **finite cardinals** are the cardinal numbers of finite sets.

The next crucial property of finite sets is the first, basic result in the field of *combinatorics*.

5.27. Pigeonhole Principle. *Every injection $f : A \rightarrowtail A$ of a finite set into itself is also a surjection, i.e., $f[A] = A$.*

PROOF. It is enough to prove that for each natural number m and each g,

$$g : [0, m) \rightarrowtail [0, m) \Longrightarrow g[[0, m)] = [0, m), \tag{5-10}$$

for the following reason: if $f : A \rightarrowtail A$ is an injection and $\pi : A \rightarrowtail\!\!\!\!\rightarrow [0, m)$ witnesses that A is finite, we define $g : [0, m) \to [0, m)$ by the equation

$$g(i) = \pi(f(\pi^{-1}(i))) \quad (i < m),$$

so that (easily)

$$f(x) = \pi^{-1}(g(\pi(x))) \quad (x \in A). \tag{5-11}$$

Now g is an injection, as a composition of injections, so that from (5-10) it is a bijection; but then f is also a bijection, because it is a composition of bijections, by (5-11).

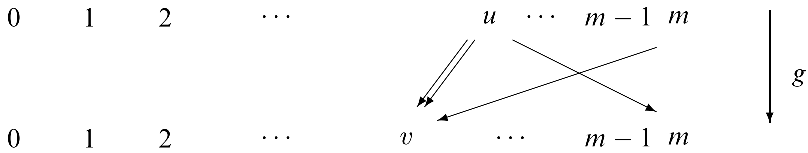

FIGURE **5.3**. CASE (3) in the Pigeonhole Principle proof.

The proof of (5-10) is (naturally) by induction on m. It is important to note at the outset that what we will show is the general assertion

$$(\forall g)\left[g : [0, m) \rightarrowtail [0, m) \Longrightarrow g[[0, m)] = [0, m)\right], \tag{5-12}$$

because in the verification of the induction step for some g we will need the induction hypothesis for various other g's.

BASIS. (5-10) is trivial when $m = 0, 1$, because there is only one function $g : [0, m) \to [0, m)$ in these cases and it is a bijection.

INDUCTION STEP. We assume (5-12) for some $m > 1$ and we proceed to prove that every injection

$$g : [0, Sm) \rightarrowtail [0, Sm)$$

is a surjection. From Exercise **5.25** we know that

$$[0, Sm) = [0, m) \cup \{m\},$$

and the proof naturally splits into three cases.

CASE (1). $m \notin \text{Image}(g)$. Consider the *restriction* h of g to the interval $[0, m)$, which is defined by

$$h(i) = g(i) \quad (i < m),$$

i.e., as a set of pairs, $h = g \setminus \{(m, g(m))\}$. This takes all its values in $[0, m)$ and it is certainly an injection, so the induction hypothesis holds for it and hence $h[[0, m)] = [0, m)$. This means that $g[[0, m)] = [0, m)$, which is absurd, because the Case Hypothesis implies $g(m) < m$ so that the value $g(m)$ is taken on twice and g is not an injection.

CASE (2). $g(m) = m$. By the same reasoning, the restriction h is a bijection $h : [0, m) \rightarrowtail\!\!\!\!\!\twoheadrightarrow [0, m)$, and hence (trivially now) g is also a bijection.

CASE (3). There exist numbers $u, v < m$ such that

$$g(u) = m, \quad g(m) = v.$$

In this most interesting case, we define the function $h' : [0, m) \to [0, m)$ by the formula

$$h'(i) = \begin{cases} g(i), & \text{if } i < m \,\&\, i \neq u, \\ v, & \text{if } i = u. \end{cases}$$

Now h' is an injection, because it agrees with g at all arguments except u, where it takes the value v; and $v \neq g(j)$, for every $j < m$, because $g(m) = v$ and g is an injection. The induction hypothesis applied to h' implies that $h'[[0, m)] = [0, m)$, and using this (easily), $g[[0, Sm)] = [0, Sm)$. ⊣

As a first application we can give a rigorous proof of the following "obvious" result.

5.28. Corollary. *The set* $\mathbb{N}$ *of natural numbers is infinite.*

PROOF. The function $(n \mapsto Sn)$ is an injection of $\mathbb{N}$ into $\mathbb{N} \setminus \{0\}$. $\dashv$

It follows that "infinite, countable" means precisely "equinumerous with $\mathbb{N}$", in accordance with our basic intuitions: a set A is countable, infinite just when $|A| =_c \aleph_0$.

5.29. Corollary. *For each finite set* A, *there exists exactly one natural number* n *such that* $A =_c [0, n)$. We let

$$\#(A) =_{\mathrm{df}} \text{the unique } n \in \mathbb{N} \text{ such that } [A =_c |A| =_c [0, n)] \tag{5-13}$$

and we naturally call $\#(A)$ *the number of elements* of A.

PROOF. If $A =_c [0, n)$ and $A =_c [0, m)$ with $n < m$, then $[0, n) =_c [0, m)$ and the correspondence $\pi : [0, m) \rightarrowtail [0, n)$ contradicts the Pigeonhole Principle, since $[0, n)$ is a proper subset of $[0, m)$. $\dashv$

From this point on we can proceed to prove all the basic properties of finite sets by induction on the number of elements in them, which is essentially their cardinal number. The method is illustrated in the problems.

5.30. Strings. In Chapter **2** we used the n-fold Cartesian product A^n to represent sequences of length n from a given set A. This is not convenient when we wish to study the set of all finite sequences from A, and it is better to represent these using functions on initial segments of $\mathbb{N}$. For each set A, we define the set of **finite sequences**, **words** or **strings** from A by

$$\begin{aligned} A^{(n)} &=_{\mathrm{df}} \{u \subseteq \mathbb{N} \times A \mid \mathrm{Function}(u) \,\&\, \mathrm{Domain}(u) = [0, n)\}, \\ A^* &=_{\mathrm{df}} \textstyle\bigcup_{n=0}^{\infty} A^{(n)}, \end{aligned} \tag{5-14}$$

and we let

$$\mathrm{lh}(u) =_{\mathrm{df}} \max\{i \mid i = 0 \vee i - 1 \in \mathrm{Domain}(u)\} \quad (u \in A^*); \tag{5-15}$$

this is the **length** of the string u, so that $\mathrm{lh}(u) = 0$ exactly when u is the empty string $\emptyset$. We also let

$$u \sqsubseteq v \iff_{\mathrm{df}} u \subseteq v \quad (u, v \in A^*), \tag{5-16}$$

and we call u an **initial segment** of v if $u \sqsubseteq v$. If $a_0, \dots, a_{n-1}$ are elements of A, we let

$$\langle a_0, \dots, a_{n-1} \rangle =_{\mathrm{df}} \{(0, a_0), \dots, (n-1, a_{n-1})\} \tag{5-17}$$

be the sequence of these objects and, in particular (with $n = 0, 1$),

$$\langle\, \rangle = \emptyset, \quad \langle a \rangle =_{\mathrm{df}} \{(0, a)\}. \tag{5-18}$$

For any two strings u, v, the string

$$u \star v = \langle u(0), \dots, u(\mathrm{lh}(u) - 1), v(0), \dots, v(\mathrm{lh}(v) - 1) \rangle \tag{5-19}$$

is the **concatenation** of the strings u and v.

For each $f : \mathbb{N} \to A$ and each natural number n,

$$\overline{f}(n) =_{\text{df}} f \upharpoonright [0, n) = \{(i, f(i)) \mid i < n\} \tag{5-20}$$

is the **restriction** of f to the initial segment $[0, n)$ of $\mathbb{N}$. For example,

$$\overline{f}(0) = \emptyset, \quad \overline{f}(1) = \{(0, f(0))\}, \dots,$$

and we can recover f from $\overline{f}$, since

$$i < n \Longrightarrow f(i) = \overline{f}(n)(i).$$

5.31. Definition. For each cardinal number κ and each $n \in \mathbb{N}$, we set

$$\kappa^n =_{\text{df}} |\kappa^{(n)}|.$$

5.32. Proposition. *For each countably infinite set A and each $n > 0$,*

$$A =_c A \times A =_c A^{(n)} =_c A^*.$$

As equations of cardinal arithmetic, these read:

$$\aleph_0 =_c \aleph_0 \cdot \aleph_0 =_c {\aleph_0}^n =_c |{\aleph_0}^*|. \tag{5-21}$$

PROOF. The inequalities from left to right are trivial, so by the Schröder-Bernstein Theorem it is enough to show $\mathbb{N}^* \leq_c \mathbb{N}$. We need to start with some injection

$$\rho : \mathbb{N} \times \mathbb{N} \rightarrowtail \mathbb{N}, \tag{5-22}$$

and let us first suppose that we have one. Using it, we define by recursion an injection $\pi_n : \mathbb{N}^{(n+1)} \rightarrowtail \mathbb{N}$, for each n, so that

$$\begin{aligned} \pi_0(u) &= u(0), \\ \pi_{n+1}(u) &= \rho(\pi_n(u \upharpoonright [0, n+1)), u(n+1)); \end{aligned}$$

in full detail (for the last time), this comes from the Recursion Theorem, by setting

$$\begin{aligned} \pi_0 &= \{(u, w) \mid u \in \mathbb{N}^{(1)}, (0, w) \in u\}, \\ \pi_{n+1} &= \{(u, w) \mid u \in \mathbb{N}^{(n+2)}, w = \rho(\pi_n(u \upharpoonright [0, n+1), u(n+1))\}. \end{aligned}$$

Finally, the function

$$\pi(u) = (\text{lh}(u) - 1, \pi_{\text{lh}(u)-1}(u))$$

proves that $\bigcup_{n=0}^{\infty} \mathbb{N}^{(n+1)} \leq_c \mathbb{N} \times \mathbb{N}$, from which the full result follows immediately by using ρ once more. As far as choosing a ρ to start with, everyone has their favorite way of coding pairs and Cantor's illustrated in Figure **2.2** will certainly do. Here is another one, due to Gödel and pictured in Figure **5.4**:

$$\rho(m, n) = \begin{cases} (m+1)^2 - 1, & \text{if } m = n, \\ n^2 + m, & \text{if } m < n, \\ m^2 + m + n, & \text{if } n < m. \end{cases}$$

The proof that it actually works is fun (Problem **x5.24**). $\dashv$

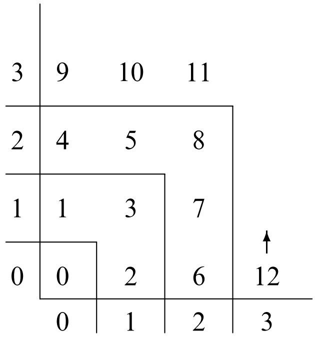

FIGURE **5.4**. A pairing of $\mathbb{N} \times \mathbb{N}$ with $\mathbb{N}$.

It is sometimes useful to think of A^* as a generalization of $\mathbb{N}$, generated from the empty string $\langle\ \rangle$ (instead of 0) by iterating the **appending operations**

$$S_a(u) = u \star \langle a \rangle \quad (a \in A),$$

one for each $a \in A$. This picture motivates the following, simple but useful method of definition by recursion on A^*.

5.33. String Recursion Theorem. *For any two sets A, E, each $a \in E$ and each function $h : E \times A \to E$, there is exactly one function $f : A^* \to E$ which satisfies the identities*

$$\begin{aligned} f(\langle\ \rangle) &= a, \\ f(u \star \langle x \rangle) &= h(f(u), x) \quad (u \in A^*, x \in A). \end{aligned}$$

Similarly, with parameters, for any

$$g : Y \to E, \quad h : E \times A \times Y \to E,$$

there exists exactly one function $f : A^* \times Y \to E$ which satisfies the identities

$$\begin{aligned} f(\langle\ \rangle, y) &= g(y) && (y \in Y), \\ f(u \star \langle x \rangle, y) &= h(f(u, y), x, y) && (u \in A^*, y \in Y, x \in A). \end{aligned}$$

PROOF. For the version with parameters, we appeal to Corollary **5.12** to obtain a function $\phi : \mathbb{N} \times A^* \times Y \to E$ which satisfies the identities

$$\begin{aligned} \phi(0, u, y) &= g(y), \\ \phi(n+1, u, y) &= h(\phi(n, u, y), u(n), y), \end{aligned}$$

and we set

$$f(u, y) = \phi(\mathrm{lh}(u), u, y).$$

Clearly $f(\langle\ \rangle, y) = \phi(0, u, y) = g(y)$, so $f(u, y)$ satisfies the first of the two required identities. To see that it also satisfies the second, we show by an easy induction on n that for any two strings u, v,

$$(\forall i < n)[u(i) = v(i)] \Longrightarrow \phi(n, u, y) = \phi(n, v, y); \tag{$*$}$$

this is immediate at the basis since $\phi(0, u, y) = g(y)$ does not depend on u, and in the induction step, assuming that $(\forall i < n + 1)[u(i) = v(i)]$,

$$\begin{aligned}\phi(n+1, u, y) &= h(\phi(n, u, y), u(n), y)\\ &= h(\phi(n, v, y), v(n), y) \quad \text{(ind. hyp. and hyp.)}\\ &= \phi(n+1, v, y).\end{aligned}$$

Now, using $(*)$, we compute, for any u with $\mathrm{lh}(u) = n$ and any $x \in A$,

$$\begin{aligned}f(u \star \langle x\rangle, y) &= \phi(n+1, u \star \langle x\rangle, y)\\ &= h(\phi(n, u \star \langle x\rangle, y), x, y)\\ &= h(\phi(n, u, y), x, y) \qquad \text{(by $(*)$)}\\ &= h(f(u, y), x, y) \qquad \text{(by the def. of f)}\end{aligned}$$

The uniqueness of $f(u, y)$ is easy, by induction on $\mathrm{lh}(u)$. ⊣

5.34. The continuum. The classical notation for the cardinal of $\mathcal{P}(\mathbb{N})$ uses the German (Fraktur) letter 'c',

$$\mathfrak{c} =_{\mathrm{df}} |\mathcal{P}(\mathbb{N})| =_c 2^{\aleph_0}. \tag{5-23}$$

It is quite easy to establish the basic facts about $\mathfrak{c}$ using the properties of $\aleph_0$ in (5-21) and elementary cardinal arithmetic. For example:

$$\mathfrak{c} \cdot \mathfrak{c} =_c 2^{\aleph_0} \cdot 2^{\aleph_0} =_c 2^{\aleph_0 + \aleph_0} =_c 2^{\aleph_0} =_c \mathfrak{c}.$$

The Schröder-Bernstein Theorem is also very useful: for example

$$\mathfrak{c} =_c 2^{\aleph_0} \leq_c {\aleph_0}^{\aleph_0} \leq_c \mathfrak{c}^{\aleph_0} =_c (2^{\aleph_0})^{\aleph_0} =_c 2^{\aleph_0 \cdot \aleph_0} =_c 2^{\aleph_0} =_c \mathfrak{c},$$

which by Schröder-Bernstein implies that

$$\mathfrak{c} =_c {\aleph_0}^{\aleph_0} =_c \mathfrak{c}^{\aleph_0}.$$

Some of the problems ask for computations of this type. On the other hand, the equinumerosity $\mathbb{R} =_c \mathcal{P}(\mathbb{N})$ will follow easily from the axioms once we have defined the real numbers $\mathbb{R}$ in Appendix **A**, so that the Continuum Hypothesis is equivalent to the proposition

$$(\mathbf{CH}) \qquad (\forall \kappa \leq_c \mathfrak{c})[\kappa \leq_c \aleph_0 \vee \kappa =_c \mathfrak{c}].$$

We will discuss **CH** extensively in Chapter **10**, it is not that easy to resolve.

Problems for Chapter 5

x5.1. Multiplication on the natural numbers is associative.

x5.2. Multiplication on the natural numbers is commutative.

x5.3. Exponentiation on the natural numbers is defined by the following recursion on m:

$$n^0 = 1,$$
$$n^{Sm} = n^m \cdot n.$$

Show that it satisfies the following identities (for $n \neq 0$):

$$n^{(m+k)} = n^m \cdot n^k,$$
$$n^{(m \cdot k)} = (n^m)^k.$$

x5.4. Suppose $(\mathbb{N}_1, 0_1, S_1)$ and $(\mathbb{N}_2, 0_2, S_2)$ are two Peano systems, $+_1$, $\cdot_1$, $+_2$, $\cdot_2$ are the functions of addition and multiplication in these systems, and $\pi : \mathbb{N}_1 \rightarrowtail\!\!\!\!\twoheadrightarrow \mathbb{N}_2$ is the "canonical" isomorphism between them defined in the proof of Theorem **5.4**. Show that π is an isomorphism with respect to addition and multiplication also, i.e., for all $n, m \in \mathbb{N}_1$,

$$\pi(n +_1 m) = \pi(n) +_2 \pi(m), \quad \pi(n \cdot_1 m) = \pi(n) \cdot_2 \pi(m).$$

x5.5. Suppose $(\mathbb{N}_1, 0_1, S_1)$ and $(\mathbb{N}_2, 0_2, S_2)$ are two Peano systems, $\leq_1$, $\leq_2$ are the respective wellorderings and $\pi : \mathbb{N}_1 \rightarrowtail\!\!\!\!\twoheadrightarrow \mathbb{N}_2$ is the canonical isomorphism. Show that π is order preserving, i.e., for all $n, m \in \mathbb{N}_1$,

$$n \leq_1 m \iff \pi(n) \leq_2 \pi(m).$$

x5.6. Every subset B of an interval $[0, n)$ is equinumerous with some $[0, m)$, where $m \leq n$. It follows that if A is finite and $B \subseteq A$, then B is finite and $\#(B) \leq \#(A)$.

Every cardinal number is a set; a **finite cardinal** κ is a cardinal number which is a finite set.

x5.7. Prove that for every finite cardinal number κ,

$$\kappa =_c [0, \#(\kappa)).$$

x5.8. Show that for all n, m, $[0, m) =_c [n, n + m)$ and infer that the union of two finite sets A, B is finite and such that

$$\text{if } A \cap B = \emptyset, \text{ then } \#(A \cup B) = \#(A) + \#(B).$$

It follows that for any two finite cardinals κ, λ,

$$\#(\kappa + \lambda) = \#(\kappa) + \#(\lambda).$$

x5.9. If $\mathscr{E}$ is a finite set and every member of it is a finite set, then the unionset $\bigcup \mathscr{E}$ is also finite.

x5.10. The product of two finite sets A, B is finite and such that

$$\#(A \times B) = \#(A) \cdot \#(B).$$

It follows that for any two finite cardinals κ, λ,

$$\#(\kappa \cdot \lambda) = \#(\kappa) \cdot \#(\lambda).$$

x5.11. The powerset of every finite set A is finite and

$$\#(\mathcal{P}(A)) = 2^{\#(A)}.$$

It follows that for every finite cardinal κ,

$$\#(2^{\kappa}) = 2^{\#(\kappa)}.$$

x5.12. For all finite cardinals κ, λ,

$$\#(\kappa^{\lambda}) = \#(\kappa)^{\#(\lambda)}.$$

x5.13. Show that if A is finite, then every surjection $f : A \twoheadrightarrow A$ is an injection. (This is an alternative version of the Pigeonhole Principle.)

x5.14. For all cardinals κ, $2^{\kappa} \neq_c \aleph_0$. (*Careful*: we have not proved the Comparability Hypothesis **3.1**, and so you cannot appeal to it.)

x5.15. $\mathfrak{c} + \mathfrak{c} =_c \aleph_0 \cdot \mathfrak{c} =_c \mathfrak{c} \cdot \mathfrak{c} =_c \mathfrak{c}$.

x5.16. $\mathfrak{c}^{\mathfrak{c}} =_c 2^{\mathfrak{c}}$.

x5.17. For every cardinal number $\kappa >_c 1$, if $\kappa \cdot \kappa =_c \kappa$, then $2^{\kappa} =_c \kappa^{\kappa}$.

x5.18. For each cardinal number κ and each $n \in \mathbb{N}$,

$$\kappa^n =_c \kappa^{|[0,n)|},$$

where the left side is defined by **5.31** and the right side is cardinal exponentiation.

x5.19. For each $n \neq 0$, $\mathfrak{c}^n =_c |\mathfrak{c}^*| =_c \mathfrak{c}$.

x5.20 (Simultaneous Recursion Theorem)**.** For any two sets E_1, E_2, elements $a_1 \in E_1$, $a_2 \in E_2$ and functions $h_1 : E_1 \times E_2 \to E_1$, $h_2 : E_1 \times E_2 \to E_2$, there exist unique functions

$$f_1 : \mathbb{N} \to E_1, \quad f_2 : \mathbb{N} \to E_2$$

which satisfy the identities

$$\begin{aligned} f_1(0) &= a_1, & f_2(0) &= a_2,\\ f_1(n+1) &= h_1(f_1(n), f_2(n)), & f_2(n+1) &= h_2(f_1(n), f_2(n)). \end{aligned}$$

***x5.21** (Nested Recursion Theorem)**.** Prove that for any three functions

$$g : Y \to E,\ h : E \times \mathbb{N} \times Y \to E, \text{ and } \pi : \mathbb{N} \times Y \to Y.$$

there is exactly one function $f : \mathbb{N} \times Y \to E$ which satisfies the identities

$$f(0, y) = g(y), \quad f(Sn, y) = h(f(n, \pi(n, y)), n, y).$$

HINT. Define recursively a function $\phi : \mathbb{N} \to (\mathbb{N} \times (\mathbb{N} \to Y))$ such that the required function is $f(n, y) = \text{Second}(\phi(n))(y)$.

x5.22. Give a recursive definition of the *factorial*

$$f(n) = 1 \cdot 2 \cdots (n-1) \cdot n \quad (\text{with } f(0) = 1, \text{ conventionally}).$$

x5.23. Give an explicit definition of the function $f : A^* \to A^*$ on the strings from some set A which is defined by the string recursion

$$f(\langle\,\rangle) = \langle\,\rangle, \quad f(u \star \langle x \rangle) = \langle x \rangle \star f(u).$$

x5.24. The function ρ in the proof of **5.32** is a bijection.

*__x5.25.__ Every partial ordering $\leq$ on a finite set P has a **linearization**, i.e., some linear ordering $\leq'$ of P exists such that $x \leq y \Longrightarrow x \leq' y$.

*__x5.26. The marriage problem__. Suppose B is a finite set and $h : B \to \mathcal{P}(G)$ is a function, such that for each $x \in B$, $h(x)$ is a finite subset of G and

$$X \subseteq B \Longrightarrow \#(X) \leq \#(\bigcup\{h(x) \mid x \in X\}), \tag{5-24}$$

so in particular each $h(x) \neq \emptyset$. Prove that there exists an injection $f : B \rightarrowtail G$ such that

$$(\forall x \in B)[f(x) \in h(x)]. \tag{5-25}$$

Show also that the hypothesis (5-24) is necessary for the existence of some injection f which satisfies (5-25). HINT. Take cases, on whether or not there exists some $\emptyset \neq C \subsetneq B$ such that $\#(C) = \#(\bigcup\{h(x) \mid x \in C\})$. The name of the problem comes from the traditional interpretation, in which B is a set of boys, G is a set of available girls and h assigns to each boy x the (finite) set $h(x)$ of girls that he would be willing to marry. There are many other applications of the problem, more useful and less sexist, for example when B is a set of courses, G is a set of professors and h assigns to each course the set of professors who can teach it ("the scheduling problem").

The next problem justifies one more form of recursive definition which is often useful in applications.

x5.27. Complete Recursion. For each $h : E^* \to E$, there exists exactly one function $f : \mathbb{N} \to E$ which satisfies the identity

$$f(n) = h(\overline{f}(n));$$

similarly, for each $h : E^* \times \mathbb{N} \to E$, there exists exactly one function $f : \mathbb{N} \to E$ which satisfies the identity

$$f(n) = h(\overline{f}(n), n).$$

The next problem gives a characterization of countable, infinite sets directly in terms of the membership relation, with no appeal to the defined notions of $\mathbb{N}$ and "function".

x5.28. Prove the equivalence:

$$\begin{aligned} A =_c \mathbb{N} \iff\ & (\exists \mathscr{E})[A = \bigcup \mathscr{E} \,\&\, \emptyset \in \mathscr{E} \,\&\, (\forall u \in \mathscr{E})(\exists! y \notin u)[u \cup \{y\} \in \mathscr{E}] \\ & \&\, (\forall Z)[[\emptyset \in Z \,\&\, (\forall u \in Z)(\exists! y \notin u)[u \cup \{y\} \in Z \cap \mathscr{E}]] \Longrightarrow \mathscr{E} \subseteq Z]]. \end{aligned}$$